

Software Requirements Specification

Francanglais Compiler and Dataset Platform

Current Requirements, Traceability and Proposed Delivery Plan

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Software completion is not evidence of fieldwork or group approval.

Revision basis: current production source in the working tree based on 160945c, including the reference dictionary extension. The two supplied SDD identities are preserved; remaining identity details, observations and historical sprint results are not inferred.

Planning basis: four proposed two-week sprints. Story points, task hours and member leads remain unconfirmed estimates, not completed work.

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1 Introduction and Document Control

1.1 Purpose, audience and revision

This SRS specifies the implemented local Francanglais/Cameroon Pidgin platform, its acceptance boundaries and remaining human deliverables. It serves contributors, the three-person project group, learners, maintainers and reviewers. The SDD explains the actual modules, all production application classes and UML workflows.

Version 1.3 retains v1.2's replacement of v1.1's desktop-only assumptions and adds the separate supplied reference dictionary. Backend, compiler and editable React source now provide direct evidence: the web API, reviewed translation, import preview, grammar workbench and coursework export are *implemented*, not hypothetical components. Existing FR-01–FR-20, NFR-01–NFR-12, US-01–US-16, PB-01–PB-16 and S1–S4 identifiers remain traceable; revised acceptance wording corrects obsolete schema and UI details. New scope receives new identifiers rather than rewriting sprint history.

1.2 Status and evidence vocabulary

Label	Meaning
B	Implemented behavior supported by current source. It is not automatically a passed manual acceptance check or a linguistically validated result.
T	An acceptance target or human/operational gate still requiring evidence. An implemented supporting mechanism does not close that gate.
D	Deferred product scope; no commitment in the four-sprint proposal.
Automated	Tests or build reports from isolated synthetic fixtures, mocks or real local handlers. A passing fixture test proves its tested behavior, not fieldwork authenticity.
Human	Genuine transcriptions, permission/method, named contributions, interpretation, device trials, provider trials and signed acceptance supplied by the group.

The reproducible verification register is in `docs/README.md` and recorded files under `docs/evidence`. Commands without an attached result are instructions, not executed evidence. No test counts, latencies, speaker findings, classifier accuracy or completed sprints are invented. The illustrative default grammar is teaching material.

1.3 Definitions

Entry

A stored word, phrase or sentence with original text, metadata and optional audio.

Gloss

A supplied French or English meaning; optional, and not guaranteed correct.

Approved

A local review flag, not an authenticated role, expert certificate or universal linguistic truth.

Grounding

Deterministic retrieval of selected approved text/gloss alignments, not model training.

LL(1)

A category-level predictive grammar using one lookahead token and an explicit end marker.

Evidence coverage

Overlap with retrieved examples; it is not a translation accuracy score.

Story point / task hour

Independent provisional planning estimates, not measured velocity.

2 Overall Description

2.1 Implemented scope

The desktop CustomTkinter collector and a FastAPI service share local UTF-8 CSV storage. React/Vite supplies five web workspaces: Translator, Learning assistant, Collection, Import & learn and Compiler lab. A deterministic Python lexer and bounded CFG/LL(1) parser run without an AI key. Exact unambiguous approved full-entry translations also run locally. Optional Gemini calls provide explicitly labelled translation/chat suggestions, media transcription and compiler explanations; they do not decide parser acceptance or approve records.

The collector has Collect, Browse & Edit and Stats tabs, optional microphone recording, attachment and playback. The backend manages reviewed alignments, validated CRUD, import previews, a three-member coursework profile, genuine user-uploaded screenshots and a locally generated draft ZIP.

2.2 Actors and operating boundary

Contributor, maintainer, learner and reviewer are usage roles, not enforced permissions. The application has **no authentication, authorization, per-user storage, durable audit trail or public-hosting hardening**. Use a trusted local workstation/loopback backend. CORS constrains browser origins; it is not access control. A remote backend's "local" files reside on that server, not automatically on the browser device.

Python and Tcl/Tk provide the desktop runtime; sounddevice, soundfile, NumPy and usable devices are optional audio dependencies. FastAPI/httpx and the compiler provide the API; React runs in the browser. Vite's development proxy forwards `/api` to the local backend. A deployed frontend requires an equivalent same-origin API arrangement, not an assumed hosting service. Browser dictation/speech synthesis are separate from desktop WAV recording.

2.3 Explicit exclusions and honest limits

No automatic fieldwork, speaker identity, consent verification, authentic corpus population, trained unseen-word predictor or certified translation accuracy is claimed. No database transaction spans CSV, audio, coursework profile and screenshots together. Cooperative file locks prevent overlapping application mutations, not manual editing races or last-writer-wins semantic conflicts. Multi-user deployment, universal grammar coverage, LR/SLR parsing, native Francanglais voices and provider retention guarantees are not supplied.

3 Functional Requirements

3.1 Retained collector and analysis requirements

ID / state	Requirement and acceptance check
FR-01 <i>B</i>	Shared storage shall initialize a missing CSV and managed audio directory without rewriting an existing valid corpus merely to read it. Acceptance: Missing storage creates a canonical header; reading 10-, 11- or 14-field supported schemas preserves existing bytes. Invalid headers/rows fail visibly rather than being silently discarded.
FR-02 <i>B</i>	Saving an expression shall reject empty or whitespace-only text. Acceptance: Desktop and API reject blank text without a row; browser validation is additional convenience, not the sole guard.
FR-03 <i>B</i>	Collection shall offer Word, Phrase and Sentence types, the ten coursework topics and Other. Acceptance: Options round-trip. Desktop defaults and reset use Word and Taxi / Commuting; web new-entry defaults are Sentence and Other.
FR-04 <i>B</i>	Contributors shall be able to supply optional French/English glosses, source location, notes and contributor text, plus explicit language/review/lexical metadata. Acceptance: Unknown provenance/meaning can remain blank. No speaker, location, approval or gloss is fabricated to complete a form.
FR-05 <i>B</i>	Creation shall generate identity and timestamp; edits preserve them. Acceptance: Desktop uses ten hexadecimal UUID characters and a local seconds-resolution ISO timestamp; API creation uses a full UUID and UTC offset. Clients cannot PATCH identity, timestamp or audio reference. No uniqueness constraint is asserted for externally written IDs.
FR-06 <i>B</i>	All CSV mutations shall produce the canonical 14-field UTF-8 schema through a locked read-modify-write transaction and atomic replacement. Acceptance: Appending, editing and deleting preserve unrelated rows and quoting; supported legacy rows gain conservative metadata on the next mutation, not during a read.
FR-07 <i>B</i>	Desktop duplicate feedback shall be advisory, debounced by 400 ms and scoped to the chosen language. Acceptance: Normalized case, whitespace, NFC and apostrophe variants warn; changing text/language clears or recomputes feedback. The desktop permits repeated observations. API duplicate rejection is separately FR-25.
FR-08 <i>B</i>	When supported, Record/Stop shall capture mono 44,100 Hz samples and stage a WAV only after successful capture and file writing. Acceptance: Mock opening/start/stop/close/write failure paths; retain retryable state, close streams and show truthful feedback. Saving while recording must not create a row. Physical sound quality remains a human test.
FR-09 <i>B</i>	A contributor shall be able to attach an existing audio file without a microphone. Acceptance: Cancellation preserves the draft; copying succeeds before replacing the pending filename; failure displays an error and preserves the previous attachment.

ID / state	Requirement and acceptance check
FR-10 <i>B</i>	Recording and attachment shall generate names inside the managed audio directory; attachment preserves the source extension. Acceptance: The source file is unchanged. Staging media does not save a CSV row. Replacement/Clear/close remove only unsaved files owned by that draft, with explicit retryable cleanup errors. Committed recordings survive draft reset and row deletion; their later cleanup is deliberate.
FR-11 <i>B</i>	Desktop Browse shall filter all stored field values case-insensitively, cache all loaded rows by ID and populate selection without a second read. Acceptance: Empty/non-match queries show all/no rows. Failed refreshes preserve usable state; stale selections are not reported as successful mutations.
FR-12 <i>B</i>	A selected desktop row shall support text, type, category, gloss, source-location, notes, language, review and lexical edits, or confirmed deletion. Acceptance: ID, timestamp, contributor and audio remain unchanged in this editor. Changed evidence resets approval unless the edit explicitly supplies a review decision. Web contributor editing is separately supported.
FR-13 <i>B</i>	Pending or selected audio shall use library playback when available, then an OS default-player fallback if necessary. Acceptance: Missing files and rejected playback/launch return failure; true means accepted playback/launch, not sound heard to completion.
FR-14 <i>B</i>	Stats shall show total rows and descending counts/bars by type, topic, language and review status from one loaded snapshot. Acceptance: Fixture totals match independent counts; empty data displays zero/no-data; normalization avoids division by zero. Failed reads preserve prior views, not an apparent empty corpus. Batch count/recent/Browse/Stats refresh shares one snapshot.
FR-15 <i>B</i>	Local analysis shall return token categories, code-mixing hints and known verb phrases; compiler-lab parsing shall return explicit acceptance/rejection with diagnostics. Acceptance: Use deterministic golden fixtures, including unsupported symbols and empty parse input. Grammar acceptance is never described as linguistic authenticity. FR-29–FR-31 refine this requirement.
FR-16 <i>B</i>	Translation retrieval, lexical/grammar analysis, parsing, chat and previews shall not change existing corpus rows or audio. Acceptance: Compare existing corpus/media hashes before/after successful and rejected requests. Missing shared storage may be initialized and lock files created; predictions are not automatically inserted.
FR-17 <i>B</i>	A successful desktop save shall refresh collection views and reset the draft while retaining contributor; Escape retains contributor and explicit Clear may remove it. Acceptance: The latest six rows are newest-first. A failed write retains draft and pending audio rather than clearing them or reporting Saved.

ID / state	Requirement and acceptance check
FR-18 <i>B</i>	Edit/delete without selection shall show feedback; deletion requires confirmation. Acceptance: No or no selection leaves storage unchanged. Yes clears selection only after a successful removal; a disappeared ID receives an actionable stale-entry error. Media files remain.
FR-19 <i>B</i>	A failed CSV mutation shall preserve the last valid dataset and report failure rather than saved/deleted success. Acceptance: Fault-inject writing and replacement: the prior CSV remains parseable and byte-identical, intermediate files are removed, and retry is possible. A later display-refresh error is not described as rollback of a committed write.
FR-20 <i>D</i>	A future unseen-word predictor shall distinguish speculative candidates from corpus facts and provide evidence or abstention. Acceptance: No training/evaluation or accuracy is claimed. Existing generative suggestions and reviewed lexical lookup do not complete this deferred predictor.

3.2 Web, reviewed grounding, imports and coursework

ID / state	Requirement and acceptance check
FR-21 <i>B</i>	The API shall expose health/configuration and collection metadata, validate request DTOs and return structured, actionable failures. Acceptance: Health reports configuration, not a live-provider probe. Unknown request fields fail validation. Document 204, 400/404/409/413/415/422, 429, 500/502/503/504 behavior and the route inventory in the SDD.
FR-22 <i>B</i>	Stored language and review status shall distinguish francanglais, pidgin, mixed and unspecified, and approved from unreviewed. Acceptance: Legacy/invalid values become unspecified/unreviewed in memory; an invalid lexical label becomes empty. Editing evidence without an explicit review decision invalidates approval. A reviewer may explicitly approve a changed record.
FR-23 <i>B</i>	Collection-grounded translation shall use approved, labelled Francanglais/Pidgin alignments and answer locally only for an unambiguous full-entry source/target match; separately enabled references follow FR-37. Acceptance: Partial/token overlap never synthesizes a local sentence. Conflicting or missing target alignments prevent local success. Return IDs, selected evidence and overlap warnings within the bounded evidence budget; do not share private row metadata.

ID / state	Requirement and acceptance check
FR-24 <i>B</i>	Explicit translation/chat requests may use configured AI, with independent collection/reference controls and visible origin/model labels. Acceptance: No eligible exact match plus disabled AI returns 422. AI output is labelled unverified, never auto-saved. Chat accepts at most six complete alternating exchanges (12 messages, 24,000 total history characters).
FR-25 <i>B</i>	The web collection shall search all row values, show whole-corpus counts, and create, partially edit or confirm deletion of entries. Acceptance: Normalized duplicate text within a language returns 409; stale IDs return 404. PATCH preserves omitted fields and rejects empty objects/null updates. UI filters and changed-field editing retain legacy values without silently rewriting them.
FR-26 <i>B</i>	Imports shall preview one supported file locally when possible, without saving rows or retaining raw imports as application records. Acceptance: TXT/MD/CSV/JSON/DOCX/text PDF work offline. Enforce 12 MiB per file, 40 PDF pages, 40,000 extracted characters, 4,000-character segments and at most 100 structured drafts. Imported approval flags cannot bypass review.
FR-27 <i>B</i>	Images, audio, video and PDFs needing OCR shall require explicit cloud-processing consent before provider submission. Acceptance: With consent off, no eligible media request reaches Gemini; a partially readable PDF may show local text plus warnings. Validate format/size; flag uncertain AI transcript text and provider retention limits.
FR-28 <i>B</i>	Import handoff and suggestions shall remain separate from saving and approval. Acceptance: Translate/Ask AI handoff only fills a browser draft. A separate suggestion request accepts only exact, word-bounded source quotations (at most 12 drafts). Each candidate opens an unreviewed editor and requires an explicit save/review decision.
FR-29 <i>B</i>	The lexer shall preserve raw tokens, apply structural regex rules first, then eligible reviewed Word annotations, static lists and suffix heuristics, ending in UNKNOWN. Acceptance: Conflicting reviewed labels are ignored; sentence rows do not train words. Unsupported characters are not dropped. Mark *_LIKE and code-mixed transitions as heuristic, not proven language identity.
FR-30 <i>B</i>	Grammar analysis shall validate the category-based notation, transform supported left recursion and common prefixes, and calculate FIRST/FOLLOW and LL(1) conflicts deterministically. Acceptance: Use explicit epsilon, preserve before/after snapshots, bound expansion, reject unsafe recursion and report remaining cycles. Conflicting table cells are absent from the executable table and retain all competing productions in diagnostics.

ID / state	Requirement and acceptance check
FR-31 <i>B</i>	A bounded table-driven parser shall consume lexer categories, requiring complete input and end-marker agreement. Acceptance: Trace stack/input before each action; distinguish missing cells, terminal mismatch, trailing tokens, conflicts and limits. UNKNOWN is a literal category, never a wildcard. Empty input can be accepted only by a suitable nullable grammar.
FR-32 <i>B</i>	Compiler lab shall maintain a locally saved profile with exactly three member strings, grammar, method, rationale, discussion, limitations and self-attestation, plus explicit screenshot CRUD. Acceptance: Missing profile uses blank members and an illustrative grammar. Save validates grammar and atomically replaces JSON under a lock. PNG/JPEG uploads are decoded/re-encoded as PNG; enforce six images, 2 MiB and 12 megapixels each.
FR-33 <i>B</i>	Corpus analysis shall use saved entries to compute frequencies, category counts, spelling-variation candidates and per-entry parse traces. Acceptance: Enforce 500 entries, 100,000 text characters and 4,000 characters per entry. Sentence target/topic counts must not be inflated by Word/Phrase rows. Readiness checks are not grades or independent fieldwork verification.
FR-34 <i>B</i>	A local export shall regenerate a coursework draft from saved profile/grammar, current saved CSV and uploaded screenshots, excluding credentials and raw audio. Acceptance: ZIP includes printable 25-section HTML, editable PPTX, CSV/JSON artifacts, selected actual sources and own-data regression snapshots. UI blocks export while project edits are unsaved. Raw text/provenance are included, so authorized review is required before sharing.
FR-35 <i>B</i>	Browser flows shall preserve drafts in tab memory, cancel stale requests, expose progress/errors and require explicit voice/submit actions. Acceptance: Edits/navigation invalidate stale output; failed mutations warn that a timeout may have committed. Dictation does not submit. Read-aloud can stop; chat auto-reading defaults off. Reload loses unsaved browser state.
FR-36 <i>B</i>	Optional compiler tutoring shall explain locally recomputed facts without applying changes or choosing parser results. Acceptance: Send only entered grammar/practice text/question/language and selected computed facts; omit stored profile, screenshots and full corpus. Bound context to 30,000 characters; suggestions remain for human review.

ID / state	Requirement and acceptance check
FR-37 <i>B</i>	The system shall expose a searchable, read-only reference dictionary from the two supplied Markdown files, separately from collected fieldwork, and use enabled reference meanings for honest local lookup. Acceptance: Preserve all 143+36 source rows, senses and source citations; support listed forms, search and pagination (at most 100 entries per API page). Require an unambiguous whole-entry match, never concatenate word meanings into a sentence. Do not invent absent French meanings, infer grammar labels from origins, write corpus rows or inflate coursework counts. Handoff fills a draft only; independent switches also constrain AI evidence. Corrupt/missing reference sources fail explicitly.

4 Non-Functional Requirements

Targets are not reported as measurements unless the verification register contains the actual environment, command and result.

ID / state	Quality requirement and verification
NFR-01 <i>B</i>	Desktop shall provide a 780 by 640 minimum window, scrollable collection controls, Ctrl+Enter Save and Escape reset. Acceptance: Exercise tabs, keyboard actions and error paths at the minimum size; physical display/usability approval remains a human check.
NFR-02 <i>B</i>	Persistence shall preserve UTF-8 and proper CSV quoting. Acceptance: Round-trip accents, commas, quotes and line breaks, including legacy schemas, without changing collected text during read normalization.
NFR-03 <i>B</i>	Audio playback shall report request acceptance separately from failure. Acceptance: Mock library/OS exceptions and missing media. Successful request return is not proof of audible playback.
NFR-04 <i>B</i>	Managed paths shall be module-relative and media references generated filenames. Acceptance: Change working directory and verify storage resolution and source-file preservation. CSV rows are not a referential-integrity database.
NFR-05 <i>B</i>	Missing optional desktop audio imports shall not prevent text collection, browsing or attachment. Acceptance: Use a dependency-unavailable fixture and device-error fixtures; real microphones and OS players need separate trials.
NFR-06 <i>B</i>	Local compiler/retrieval results shall be deterministic for fixed inputs, corpus, grammar and rules. Acceptance: Compare repeated structured results in golden fixtures. This does not promise deterministic generative-provider wording.

ID / state	Quality requirement and verification
NFR-07 <i>B/T</i>	Critical software paths shall have isolated automated regression coverage and an explicit evidence record. Acceptance: Run the Python, frontend unit and browser suites documented in the verification register; never use or mutate the real corpus as test data. Human release acceptance remains pending.
NFR-08 <i>B/T</i>	Secrets and personal material shall have explicit publication/privacy boundaries. Acceptance: Server-only secret configuration, selected evidence payloads, opt-in cloud upload and repository exclusions are implemented. Permission to collect/share, authorized export review and any old-history cleanup require humans; ignoring a file is not retroactive deletion.
NFR-09 <i>B</i>	LaTeX and every PlantUML source shall build locally into readable, consistently referenced documentation. Acceptance: Render all diagrams with the local jar, compile both PDFs, validate full class inventory, balanced communications and references; inspect layout and remaining human placeholders.
NFR-10 <i>B/T</i>	On a recorded reference Windows workstation with 8 GB RAM and local SSD, search and Stats refresh shall each have p95 latency at most one second for 1,000 rows of up to 500 text characters. Acceptance: Use one warm-up and 20 measured active-tab handler runs with Tk idle rendering and nearest-rank p95. Evidence E4 passes the unchanged timing threshold on a 31.8 GiB machine. It does not certify the original 8 GB/SSD configuration; that separate acceptance gate remains open.
NFR-11 <i>B</i>	CSV mutations shall be failure-safe and cooperative across processes, with truthful desktop/API error feedback. Acceptance: A reentrant path-keyed FileLock with 10-second timeout encloses complete transactions; same-directory write, flush/fsync and replace protect the prior file. Fault tests do not prove every filesystem or power-loss scenario.
NFR-12 <i>B/T</i>	A safe offline backup/verify/restore procedure shall cover the corpus, referenced audio and local coursework evidence. Acceptance: Use the snapshot CLI in the SDD/README, restore only into a new directory, and compare manifest hashes. A synthetic rehearsal verifies tooling, not an authorized backup of authentic fieldwork.
NFR-13 <i>B/T</i>	The backend shall be treated as a trusted local service, not a publicly secure multi-user system. Acceptance: Document no authentication, authorization, rate-limiter or account separation; keep keys server-side and limit network exposure. Public deployment needs separate design and review.

ID / state	Quality requirement and verification
NFR-14 <i>B</i>	Untrusted inputs and provider calls shall have bounded work and safe errors. Acceptance: Check DTO lengths, file and grammar limits, parser progress guards, provider timeout and one 503 retry. Do not expose provider bodies or secret-bearing diagnostics to clients.
NFR-15 <i>B/T</i>	Web pages shall provide labelled controls, focus-managed dialogs, keyboard paths, responsive layout and visible error/status feedback. Acceptance: Automated browser/DOM checks cover implemented flows; real assistive-technology, mobile-device and pronunciation acceptance remain open.
NFR-16 <i>B/T</i>	Human evidence shall remain distinguishable from automated output and self-attestation. Acceptance: Label imported/AI text, illustrative grammar, generated expected tests and draft reports. Supply genuine fieldwork, group identities/contributions, original interpretation and final review before submission.

5 Data and External Interface Requirements

5.1 Canonical storage and compatibility

The header order is part of the contract. The SDD specifies the actual locks and transaction boundary; the following is the logical schema, not an invented persistence model class.

Field (in order)	Meaning
<code>id</code>	Generated stable identifier; historical desktop and full API UUID formats coexist.
<code>text</code>	Collected wording. The API preserves stored strings; a new browser/desktop form trims surrounding whitespace.
<code>entry_type</code>	Word, Phrase or Sentence; legacy values can remain on an unchanged row.
<code>french_gloss,</code> <code>english_gloss</code> <code>category</code>	Optional alignments; no automatic correctness guarantee.
<code>source_location</code>	Required topic selection on new UI/API entries; legacy values are retained where unchanged.
<code>notes</code>	Optional provenance/location, not a geolocation sensor or proof of permission.
<code>audio_filename</code>	Optional context, uncertainty and private review notes.
<code>contributor</code>	Optional one-file reference; no file existence/ownership uniqueness constraint.
<code>timestamp</code>	Optional supplied text, not an account or mandatory matricule field.
<code>language</code>	Creation timestamp; desktop local and API UTC formats remain supported.
	francanglais, pidgin, mixed or unspecified. Unknown/legacy defaults to unspecified.

Field (in order)	Meaning
<code>review_status</code>	unreviewed or approved. Unknown/legacy defaults to unreviewed.
<code>lexical_category</code>	Empty or one of the eleven declared lexer terminals.

The 11-field pre-review schema is the canonical prefix ending at timestamp. The 10-field legacy schema additionally omits source location. Exact header order is validated; arbitrary columns and incomplete/extra cells are rejected. Reads upgrade metadata in memory. Mutations write all fourteen fields atomically.

Topics are Taxi / Commuting, Internet Connectivity, Electricity Supply, Market Bargaining, Rainy Season, Fuel Scarcity, Roadside Business, Bendskin Communication, Security Checkpoint and Campus Life, plus Other. They are *collection categories*, not lexer terminal labels.

5.2 Interface families

All explicit application routes use `/api`: health/metadata, translation, chat, lexical analysis, dataset CRUD, read-only dictionary lookup, import preview/suggestions, and coursework state, project saving, corpus analysis, parsing, tutoring, screenshots and export. DTO details and the complete method/path table are in the SDD. Request models reject unknown JSON fields. The OpenAPI page is generated by FastAPI; it is documentation, not proof that an AI key is valid. Desktop has no HTTP dependency.

5.3 Privacy and evidence separation

Local retrieval sends ID, expression text, language, French/English glosses, match type and a source discriminator, with coverage where applicable. Dictionary matches additionally include their supplied forms and reference filename/line; these are not human fieldwork claims. Collection source location, contributor, notes, timestamp, audio and full corpus are excluded. This minimization does not anonymize sensitive words a user explicitly submits. Cloud consent applies to import preview; pressing a later AI suggestion, translation, chat or tutoring action is a separate submission. Browser speech providers may receive audio independently of Gemini. The coursework export *does include* corpus text, contributor/provenance and the saved project; it must be reviewed before sharing. These documentation PDFs contain neither the actual corpus nor invented fieldwork.

6 User Stories and Acceptance Criteria

US-01–US-16 preserve the previous story identities. US-17–US-24 describe already implemented added scope. Each maps to the corresponding PB suffix. None of these statements records a historically completed sprint.

6.1 US-01 — Initialize a usable corpus

As a contributor, I want safe initialization so I can resume collection without losing existing entries.

Trace: FR-01, FR-06; NFR-02, NFR-04.

Acceptance criteria

- Missing storage gets a canonical header; supported existing storage survives a read byte-for-byte.
- Corrupt or unsupported storage fails visibly instead of being replaced with an empty corpus.

6.2 US-02 — Capture an annotated example

As a contributor, I want exact text with optional meanings and context so that the record retains useful evidence.

Trace: FR-02–FR-06, FR-17, FR-22.

Acceptance criteria

- a) Blank text is rejected; valid fields, generated metadata and language/review decisions round-trip.
- b) Only a committed save resets the draft; genuine unknown facts can remain blank.

6.3 US-03 — Notice duplicate text

As a desktop contributor, I want advisory duplicate feedback so that legitimate repeated observations remain possible.

Trace: FR-07; FR-25 for the different web policy.

Acceptance criteria

- a) Matching normalization includes case, whitespace, apostrophes and language.
- b) The warning is debounced and non-blocking; web duplicates instead return 409.

6.4 US-04 — Find and select an entry

As a maintainer, I want search and a reliable selection so I can locate annotations for review.

Trace: FR-11, FR-25; NFR-10.

Acceptance criteria

- a) Desktop empty/non-matching queries show all/no rows; selection uses the cached row.
- b) Web filters apply to returned rows, while displayed aggregate counts remain whole-corpus counts.

6.5 US-05 — Correct an annotation

As a maintainer, I want controlled editing so that identity and omitted evidence stay intact.

Trace: FR-06, FR-12, FR-18, FR-22, FR-25.

Acceptance criteria

- a) Invalid/stale/no-selection edits do not claim success.
- b) Changed evidence invalidates approval unless review is explicit; identity, timestamp and media are preserved.

6.6 US-06 — Delete with confirmation

As a maintainer, I want a confirmation boundary to avoid accidental removal.

Trace: FR-12, FR-18, FR-25.

Acceptance criteria

- a) No/cancel changes nothing; Yes removes the matching row only after a successful transaction.
- b) Disappeared rows report an error; audio remains for deliberate separate cleanup.

6.7 US-07 — Attach existing audio

As a contributor, I want to stage an existing file without a microphone.

Trace: FR-09, FR-10; NFR-04, NFR-05.

Acceptance criteria

- a) Success copies to a generated filename; cancellation/failure preserves the previous draft attachment.
- b) Saving links the staged name, without modifying the original file.

6.8 US-08 — Listen to an example

As a contributor, I want playback feedback so I can check the linked recording.

Trace: FR-13; NFR-03.

Acceptance criteria

- a) Library failure may reach the OS fallback; missing media or rejected fallback shows an error.
- b) Playback does not mutate CSV; physical audibility is not inferred from a mocked success.

6.9 US-09 — Record an example

As a contributor, I want a recoverable Record/Stop lifecycle so that captured pronunciation can accompany an entry.

Trace: FR-08, FR-10, FR-19; NFR-05, NFR-11.

Acceptance criteria

- a) A successful capture stages mono 44,100 Hz WAV; no frames create no new recording.
- b) Save while recording is blocked; start/stop/write errors leave truthful retryable state.

6.10 US-10 — Review corpus coverage

As a maintainer, I want counts by type/topic so I can find gaps.

Trace: FR-14, FR-33.

Acceptance criteria

- a) Known fixtures match independent counts; empty data is safe.
- b) Sentence/topic readiness is not inferred from Word/Phrase counts or from a chart alone.

6.11 US-11 — Protect corpus quality and privacy

As a maintainer, I want safe mutations, regression checks and recovery so that evidence remains usable and controlled.

Trace: FR-19; NFR-07, NFR-08, NFR-10–NFR-14.

Acceptance criteria

- a) Fault tests preserve prior data; tests/benchmarks use synthetic storage.
- b) Verify and restore a snapshot into a new destination, with matching hashes; sharing still needs approval.

6.12 US-12 — Build understandable documentation

As a group member, I want explained editable diagrams and locally built documents.

Trace: NFR-09, NFR-16.

Acceptance criteria

- a) Every production class has an inventory entry and diagram; every diagram has an explanation and trace.
- b) Local rendering/build leaves no unresolved references or overflowing content; human placeholders remain explicit.

6.13 US-13 — Inspect lexical analysis

As a learner, I want visible tokens and category evidence so that I can inspect the lexer's limits.

Trace: FR-15, FR-16, FR-29; NFR-06.

Acceptance criteria

- a) Raw tokens survive segmentation; approved Word annotations affect only eligible non-conflicting entries.
- b) Unsupported symbols become UNKNOWN; heuristic categories and inferred transitions are labelled as such.

6.14 US-14 — Inspect sentence analysis

As a learner, I want transformations, sets, table cells and traces so that I can explain a declared grammar.

Trace: FR-15, FR-16, FR-30, FR-31; NFR-06, NFR-14.

Acceptance criteria

- a) Valid fixtures have deterministic results; malformed, conflicting and bounded-out cases have explicit diagnostics.
- b) The illustrative starter and category-level acceptance are never claimed as a validated corpus grammar.

6.15 US-15 — Receive a demonstrable release

As a reviewer, I want repeatable software evidence separated from unfinished research deliverables.

Trace: FR-19, FR-34; NFR-07–NFR-16.

Acceptance criteria

- a) Handover contains built documents, actual verification reports, recovery instructions and known limits.
- b) Group review, authentic fieldwork, device/provider trials and final manuscript/presentation remain open until evidenced.

6.16 US-16 — Explore unseen-word candidates

As a learner, I may want future speculative candidates without confusing guesses with corpus facts.

Trace: FR-20, FR-16.

Acceptance criteria

- a) The trained/evaluated predictor is deferred; no automatic insertion or unevaluated accuracy claim is permitted.

6.17 US-17 — Use the local web collection

As a web contributor, I want validated CRUD and metadata so that I can maintain shared local entries.

Trace: FR-21, FR-25, FR-35.

Acceptance criteria

- a) Search, filtering, creation, changed-field review and confirmed deletion return the expected data and HTTP status.
- b) Rejected/ambiguous mutations preserve drafts and prompt refresh rather than blind retries.

6.18 US-18 — Translate from reviewed alignments

As a learner, I want exact reviewed local translations and honest coverage when evidence is incomplete.

Trace: FR-22, FR-23, FR-24.

Acceptance criteria

- a) Exact unambiguous alignments need no AI key; phrase/token overlap is not a composed local translation.
- b) Disabled AI with no exact answer reports 422; evidence, origin and warnings remain visible.

6.19 US-19 — Request optional language assistance

As a learner, I want explicit AI suggestions with controllable context and no automatic learning.

Trace: FR-24, FR-35, FR-36; NFR-08, NFR-14.

Acceptance criteria

- a) Translation/chat/tutoring send only the declared input and allowed evidence; output is unverified.
- b) Failed chat turns do not enter successful exchange history; clear/navigation cancels stale work.

6.20 US-20 — Preview, review and import material

As a contributor, I want local extraction or consented transcription before deciding what to save.

Trace: FR-26–FR-28.

Acceptance criteria

- a) Limits, unsupported formats, corrupt input and no-consent cases fail explicitly.
- b) Preview, handoff and suggestions save nothing; each reviewed candidate is saved separately with truthful provenance.

6.21 US-21 — Maintain coursework evidence

As a group member, I want a saved profile, corpus analysis and genuine screenshots without generated research claims.

Trace: FR-32, FR-33; NFR-16.

Acceptance criteria

- a) Exactly three distinct nonblank member strings can be supplied; blanks remain visibly incomplete.
- b) Editor grammar can be analyzed without saving. Saved evidence refresh preserves editor drafts; self-attestation is not independent validation.

6.22 US-22 — Export a coursework draft

As a group member, I want a locally reproducible bundle derived only from saved evidence.

Trace: FR-34; NFR-08, NFR-12, NFR-16.

Acceptance criteria

- a) Dirty editor state blocks export until save. ZIP uses saved grammar/CSV/screenshots, not AI replies or unsaved practice text.
- b) Review report pagination, slide timing, generated expectations and sensitive content before authorized submission.

6.23 US-23 — Control browser requests and voice

As a browser user, I want accessible, explicit actions that do not submit stale or unintended content.

Trace: FR-35; NFR-15.

Acceptance criteria

- a) Cancellation/navigation/editing prevents stale success callbacks; timeouts never promise server rollback.
- b) Dictation is opt-in and only fills text; synthesis can stop and has no native-voice guarantee.

6.24 US-24 — Use supplied reference vocabulary without inventing fieldwork

As a learner, I want to find supplied words and English meanings even when the research CSV is empty, while keeping their provenance distinct.

Trace: FR-23, FR-24, FR-35, FR-37; NFR-08, NFR-16.

Acceptance criteria

- a) Both supplied sources are searchable and cited; conflicting senses remain visible and never silently replace each other.
- b) Exact reference lookups need no AI key; missing French remains a gap. The reference switch is independent of collection use.
- c) Dictionary handoff does not submit/save; collection counts and coursework corpus analysis remain unchanged.

7 Prioritized Product Backlog

7.1 Current verification register

These results describe this source snapshot, not an invented sprint history. E5 is verified by rebuilding and checking the published artifacts; the same commands run in the configured CI workflow. A remote CI result is not claimed.

ID	Recorded result and boundary
E1	268 Python regressions passed, including 16 dictionary, 56 desktop and 33 maintenance/checker tests. Earlier real WAV codec/file round-trip also passed; microphone/provider tests remain isolated mocks.
E2	86 frontend component tests passed; application and test TypeScript checks and production build passed.

ID	Recorded result and boundary
E3	22 desktop/mobile Chromium cases passed, including dictionary search/handoff/retry, import consent, review, keyboard focus, saved-project/export gating and downloaded fixture ZIP bytes. API data is mocked, not authentic research.
E4	Real active-tab benchmark: 1,000 synthetic 500-character rows; search p95 53.31 ms, Stats p95 606.01 ms; 20 measured runs each. Report: <code>docs/evidence/collector-benchmark.json</code> . Measured hardware is not the original 8 GB/SSD reference.
E5	SRS/SDD and 29 full-size UML sheets built locally. Inventory covers 49 production classes; 14 sequence views have explicit balanced call/reply activations. Source/sheet/image/reference checks accompany publication.
E6	Reference extension: both supplied files retain 143+36 source entries, explicit English meanings and provenance. Local lookups, conflicting senses, missing French, source switches and unchanged corpus/export counts are tested. See <code>docs/evidence/dictionary-verification.json</code> .
H	Human gates: genuine corpus and permission, remaining identity details and all contributions, reviewed linguistic findings, real devices/provider, reference hardware, final report/slides and group sign-off. Not fulfilled by E1–E6.

7.2 Source state versus proposed scheduling

The original point estimates and proposed sprint selections are retained for traceability: 67 selected SP, plus 8 deferred SP, originally 75 total. These are **not remaining-effort estimates**. Current implementation is ahead of that old proposal in several areas; the group must re-estimate and agree a real schedule before using it. “Implemented” below is not “sprint Done” and is not group sign-off.

PB	US	Priority	SP	Plan	Depends	Current state / remaining evidence
PB-01	US-01	Must	3	S1	None	Software verified (E1): initialization, strict schemas and conservative legacy reads.
PB-02	US-02	Must	5	S1	PB-01	Software verified (E1–E3): annotated creation/reset; authentic content remains H.
PB-03	US-03	Should	3	S1	PB-02	Software verified (E1): advisory desktop duplicates; web 409 policy is distinct.
PB-04	US-04	Must	5	S1	PB-01	Software verified (E1, E3); measured timing E4. Original reference-machine gate remains H.
PB-05	US-05	Must	5	S2	PB-04	Software verified (E1–E3): changed fields, review reset, stale IDs and failures.
PB-06	US-06	Must	3	S2	PB-04	Software verified (E1, E3): confirmation/removal; committed audio retained.
PB-07	US-07	Should	3	S2	PB-02	Software verified (E1): attachment, cancellation and draft-file ownership.
PB-08	US-08	Should	5	S2	PB-07	Software verified (E1): fallback/errors. Real audibility/device acceptance is H.
PB-09	US-09	Should	5	S3	PB-07,08	Software/WAV I/O verified (E1): retry states and guards. Genuine microphone trial is H.
PB-10	US-10	Should	3	S3	PB-01	Software verified (E1, E4): consistent Stats/readiness. Authentic topic coverage is H.

PB	US	Priority	SP	Plan	Depends	Current state / remaining evidence
PB-11	US-11	Must	5	S3	PB-05–09	Software verified (E1, E4): fault safety and restore checks. Real-data permission/recovery approval is H.
PB-12	US-12	Must	3	S3	PB-11	Engineering documents verified (E5); remaining member details and group approval remain H.
PB-13	US-13	Must	8	S4	PB-01,11	Software verified (E1): regex/reviewed-word lexer. Independent linguistic evaluation is H.
PB-14	US-14	Must	8	S4	PB-13	Software verified (E1, E3): CFG/LL(1)/traces. Own-corpus rationale/evaluation is H.
PB-15	US-15	Must	3	S4	PB-12–14	Engineering handover verified (E1–E5); research, devices, provider and group sign-off remain H.
PB-16	US-16	Could	8	Later	PB-13,14	Deferred trained/evaluated unseen-word predictor; generative suggestions are not its completion.

7.3 Added implemented scope, not retroactively assigned to sprints

No story points, task hours or sprint dates are invented for these additions. They require group prioritization/re-estimation, although the software exists.

Item / story	Depends on	State and next acceptance gate
PB-17 / US-17	PB-01,05,06	Software verified (E1–E3): web CRUD/metadata and failure handling.
PB-18 / US-18	PB-13,17	Software verified (E1–E3): exact retrieval/coverage. Genuine reviewed alignments remain H.
PB-19 / US-19	PB-18	Software verified with mocked provider (E1–E2). Authorized live service/quality trials remain H.
PB-20 / US-20	PB-17,19	Software verified (E1–E3): bounded preview, consent, handoff and explicit save. Authentic source/permission remain H.
PB-21 / US-21	PB-13,14,17	Software verified (E1–E3): profile/analysis/images. Actual group data, transcripts and interpretation remain H.
PB-22 / US-22	PB-21	Software verified (E1–E3): draft ZIP/report/slides and native download. Final pagination/rehearsal/release remain H.
PB-23 / US-23	PB-17–22	Software verified (E2–E3): cancellation, focus and opt-in controls. Real devices/assistive technology remain H.
PB-24 / US-24	PB-18,23	Software verified (E1–E3, E6): separate supplied dictionary, provenance-aware retrieval and draft handoff. Source coverage and linguistic evaluation remain human responsibilities.

8 Sprint Backlogs: Four Proposed Two-Week Iterations

8.1 Planning assumptions and unconfirmed leads

S1–S4 remain a **proposal**, not retrospective history. Relative weeks 1–8 have no claimed calendar dates, completed ceremonies or measured velocity. M1, M2 and M3 refer to the three author rows; suggested leads are **unconfirmed**. All tasks below have planning allocation state **Proposed**. Each task separately records current software evidence below; it does not claim a completed sprint.

Implementations should be reviewed and verified instead of unnecessarily re-created. Added PB-17–PB-24 scope is not silently included in these unchanged estimates.

Sprint	Window	Proposed goal	SP	Hours
S1	Weeks 1–2	Reliable collection/discovery acceptance	16	24
S2	Weeks 3–4	Controlled maintenance and audio-file acceptance	16	26
S3	Weeks 5–6	Recording, failure safety and documentation evidence	16	29
S4	Weeks 7–8	Local compiler and honest handover acceptance	19	27

8.2 Sprint 1 proposal — Collection and discovery

Selected: PB-01–PB-04, 16 SP / 24 estimated hours.

Task	PB	Task and proposed evidence	Lead	h
S1-T01	PB-01	Review all supported headers and isolated missing/existing-storage fixtures. Current evidence: Verified E1.	M1	3
S1-T02	PB-01	Check UTF-8, quoting, strict read failures and non-destructive startup. Current evidence: Verified E1.	M3	3
S1-T03	PB-02	Review fields, blank validation, review defaults and generated metadata. Current evidence: Verified E1–E3.	M2	4
S1-T04	PB-02	Verify save/reload, latest rows, retained contributor and shortcuts. Current evidence: Handlers verified E1; physical UI review H.	M3	3
S1-T05	PB-03	Review debounce, language scope and matching normalization. Current evidence: Verified E1.	M1	2
S1-T06	PB-03	Test advisory appearance/clearance and permitted desktop duplicates. Current evidence: Verified E1.	M3	2
S1-T07	PB-04	Review filter/cache selection and representative synthetic fixtures. Current evidence: Verified E1, E3.	M2	4
S1-T08	PB-04	Test query edge cases and record actual handler timings. Current evidence: Measured E4; original hardware gate H.	M3	3

Proposed exit: demonstrate valid/blank save, advisory duplicate and matching search. Attach evidence; do not mark a sprint completed from code alone.

8.3 Sprint 2 proposal — Maintenance and audio files

Selected: PB-05–PB-08, 16 SP / 26 estimated hours. Depends on S1.

Task	PB	Task and proposed evidence	Lead	h
S2-T01	PB-05	Review allowed fields, review invalidation and locked mutation. Current evidence: Verified E1–E3.	M1	4
S2-T02	PB-05	Test valid/blank/no-selection/stale edits and preserved metadata. Current evidence: Verified E1–E3.	M3	4
S2-T03	PB-06	Review confirmation, ID filtering and post-success state cleanup. Current evidence: Verified E1, E3.	M2	3
S2-T04	PB-06	Test Yes/No, stale rows and unchanged audio after deletion. Current evidence: Verified E1, E3.	M3	3
S2-T05	PB-07	Review chooser cancellation, naming and microphone-free copying. Current evidence: Verified E1.	M1	3
S2-T06	PB-07	Test original-file preservation and failed attachment state. Current evidence: Verified E1.	M3	3
S2-T07	PB-08	Review playback and platform fallback boundaries. Current evidence: Software verified E1; physical sound H.	M2	3
S2-T08	PB-08	Mock missing media/library/launch errors; plan genuine device trial. Current evidence: Mocks verified E1; real trial H.	M3	3

Proposed exit: show editing, confirmed/cancelled deletion, attachment, playback-error feedback and no unwanted media deletion.

8.4 Sprint 3 proposal — Quality, recording and documentation

Selected: PB-09–PB-12, 16 SP / 29 estimated hours. These mechanisms now exist; this proposal reviews their evidence rather than claiming history.

Task	PB	Task and proposed evidence	Lead	h
S3-T01	PB-09	Review stream lifecycle, save guard and device-failure recovery. Current evidence: Software verified E1; real device H.	M1	3
S3-T02	PB-09	Test frames/no frames, unavailable audio and start/stop/write failures. Current evidence: Verified E1, including real WAV I/O.	M3	4
S3-T03	PB-10	Review grouping, totals and descending normalized bars. Current evidence: Verified E1, E4.	M2	3
S3-T04	PB-10	Test empty corpus, known counts and refresh failure behavior. Current evidence: Verified E1.	M3	2

Task	PB	Task and proposed evidence	Lead	h
S3-T05	PB-11	Verify cross-process locking, atomic replacement and truthful write errors. Current evidence: Verified E1.	M1	6
S3-T06	PB-11	Review regression/benchmark/snapshot evidence and consent/exclusions. Current evidence: Tooling verified E1, E4; permission H.	M3	4
S3-T07	PB-12	Reconcile all UML, classes, requirements and local build steps. Current evidence: Engineering evidence E5; group sign-off H.	M2	4
S3-T08	PB-12	Render diagrams, compile PDFs and inspect readability/traceability. Current evidence: Published build and checks E5.	M3	3

Proposed exit: evidence of recording behavior, Stats, fault safety, an isolated restore rehearsal and reproducible documents. Physical-device approval and collection permission cannot be supplied by automated fixtures.

8.5 Sprint 4 proposal — Compiler and handover

Selected: PB-13–PB-15, 19 SP / 27 estimated hours. A real group must approve its fixture corpus and grammar rationale before making linguistic claims.

Task	PB	Task and proposed evidence	Lead	h
S4-T01	PB-13	Review deterministic tokenization, annotations and heuristic boundaries. Current evidence: Software verified E1; linguistic review H.	M1	6
S4-T02	PB-13	Check mixed-input golden tests, UNKNOWN and read-only hashes. Current evidence: Verified E1.	M3	4
S4-T03	PB-14	Review CFG transformations, FIRST/FOLLOW, LL(1) and provenance. Current evidence: Software verified E1; own-corpus rationale H.	M2	6
S4-T04	PB-14	Test accepted/rejected/empty/conflicting/limited cases and repeatability. Current evidence: Verified E1, E3.	M3	4
S4-T05	PB-15	Review end-to-end software evidence, privacy, recovery and measured performance. Current evidence: Verified E1–E4; live/hardware gates H.	M1	4
S4-T06	PB-15	Assemble documents, actual evidence, limitations and the group’s own demo. Current evidence: Engineering artifacts E1–E5; genuine demo H.	M2	3

Proposed exit: traceable acceptance evidence and honest open gates. PB-16 remains de-

ferred. The four proposals retain **106 estimated task hours**; no team availability or historical completion is inferred.

8.6 Ready, Done and change control

Ready: agree actor value, acceptance criteria, dependencies, fixture/permission availability, estimate and an actual member lead. **Done:** reviewed implementation, applicable recorded tests, current documentation and no unresolved critical data-loss/privacy issue in that agreed scope. Human acceptance conditions also need genuine evidence. Use Proposed, In progress, Blocked and Done on an actual board, with dates and owners only when supplied. Replan added scope before assigning it to an iteration. Source-supported functionality is never retroactively a sprint record.

9 Traceability, Verification and Remaining Human Inputs

9.1 Design and verification map

Stories / PB	SDD views	Evidence boundary
01–03	Collection activity/save; storage sequence	Initialization/schema/CRUD fixtures and real-handler tests; real observations still human.
04–06	Search/edit/delete; entry state	Cached selection, invalid/stale cases, confirmation and fault-injection tests.
07–09	Recording/playback; collector classes	Mocked stream/files/OS outcomes; actual microphones and audibility separate.
10–12	Stats, component/deployment, class inventory	Synthetic counts/benchmark, safe restore, local rendering/PDF validation.
13–14	Compiler classes; grammar/parse sequence	Golden transformations, sets, conflicts, guards and trace tests; no linguistic accuracy inference.
15–16	Limitations and handover	Software evidence versus pending sign-off; predictor remains deferred.
17–19	API DTO/provider classes; translation/chat sequences	Request validation, privacy payloads, local exact path and mock-provider errors.
20	Import/coursework DTOs; import/review sequence	Local extraction/limits/no-save and explicit consent/review tests.
21–22	Coursework save/export sequence	Saved snapshot, image limits, ZIP contents and draft warnings; actual final report/slides need review.
23	Frontend class/modules; component view	Unit/browser cancellation, navigation, dialogs, errors and voice mocks; real assistive technology/device checks open.

9.2 Required fieldwork and group input checklist

1. **Exact statements:** provide 10–15 genuinely heard, manually transcribed sentence records, preserving wording, accents, code mixing, ellipsis and uncertainty. Map them to the ten

topics. Demo prompts, AI translations, OCR and machine transcripts are not substitutes for listening.

2. **Provenance and meanings:** supply collection context/location, collector and dates where genuinely known and permitted; record uncertainty. Provide French/English meanings only when known or reviewed; otherwise leave them blank. Use mixed/unspecified language labels when appropriate.
3. **Method and permission:** describe how/when/where collection occurred, permission to record/process/share, anonymization and any restrictions. Keep sensitive permission evidence in an appropriately restricted place, not automatically in a public repository.
4. **Three actual members:** the first author's provided name/matricule are already retained on the title pages. Supply the other two identities and all individual contributions to collection, implementation, analysis, writing and presentation. The profile has three free-text member slots, not separate verified identity fields. Complete document title pages and contribution descriptions truthfully.
5. **Original interpretation:** write the group's grammar rationale, findings on lexical variation/code mixing, discussion, limitations and citations; link claims to collected IDs. Review parser rejections instead of calling the speech wrong. Generated test expectations need independent review.
6. **Genuine demonstrations:** supply screenshots of actual work, group acceptance, physical microphone/player and browser/device observations, and separately authorized live-provider checks. Mark self-attestation only when true; software cannot establish authenticity.
7. **Final submission:** review the actual coursework manuscript for 25–30 pages (maximum 30) and rehearse the 10-minute presentation, approximately three minutes per member plus shared wrap-up. These SRS/SDD PDFs are supporting engineering documents, not the required research report.

9.3 Release boundary

Software, documentation and synthetic verification can be complete while the human submission remains incomplete. Remaining gates must retain their actual status: no authentic corpus, group sign-off, physical-device success, provider quality result, completed sprint or measured accuracy is filled in merely because tests/builds pass.